

27/05/2026

Assignment-1

1. what is the output of the following and why?

```
S = python  
print(S[0], S[-1], S[6])
```

output:

The output will be the Index error.
because. In the string the Index value is
from 0 to 5 only so S[6] is out of range

2. what is the difference between positive and negative Indexing

The positive Index will go straight order
which is called traversing.

The negative Index will be in reverse
order (back traversing)

Given

S = 'Hello'

print(S[-2])

output

l

3. what happens when you access a Index that is out of range. In a string.

Ans:

Index error will be occur when
it is out of range.

How it is different from slicing out of range?

`s = 'code'`

`print(s[10])` vs `print(s[1:10])`

Ans

When we print `(s[10])` the index value 10 alone will be printed. here `(s[10])` is the out of range so index error will be occur.

For `print(s[1:10])` it start print the index value from 1 to 9. but here also index error will occur because the index value for string is from 0 to 3 only.

Q. How do you find the index of the first and last occurrence of a character in the string using indexing concept.

given `s = "mississippi"`

we can find the index value of first and last using the left index and right index concept.

output-

`print(s.lindex('s'))`

2

`print(s.rindex('s'))`

6

5. what is the output of the following, and explain how negative index solve.

```
s = 'abcdef'
```

```
print(s[-1], s[-3], s[-6])
```

output:

f d a

because here we using back traversing, for negative index it goes from in reverse order of the string value

6. Syntax for string slicing in python. Explain each part.

Syntax

String [start: stop: step]

start = start is the starting value of the index number. It default is '0'

stop: Ending of the index value in the string

step: Increment value of the index at the string.

7. What is the output of the following slices

```
s = 'Hello world'
```

```
print(s[0:5])
```

Hello

```
print(s[5:])
```

World

```
print(s[:5])
```

Hello

```
print(s[:])
```

Hello world

8. What happens when start is greater than stop in a slice in positive step.

Ans:

If the start index value is greater than end value. It won't print anything as the output.

9. How do you extract the last-3 characters of a string using slicing, without knowing its length?

We can use `s[-3:]`.

example:

```
s = 'kishore'
```

```
print(s[-3:])
```

Output:

ore

10. what is the difference between this slice given,

`s = 'abcdefgh'`

`print(s[:])` → here default step is there so it prints the whole string
a b c d e f g h

`print(s[::2])` → here step value is 2 so it leaves in between, one value and prints
a c e g

`print(s[1::2])` → here first value should skip and start printing from the 2 value.
b d f h

11. How do you reverse the string using slicing.

Ans:

The step value is -1 to reverse the order.

Ex

`s = 'python'`
`print(s[::-1])`

output:

n o t h y p

12. what is the output of the following

`s = 'abcdefgh'`

`print(s[::-1])`

h g f e d c b a

`print(s[::2])`

g e c a

`print(s[6:1:-1])`

→ It will reverse the order

→ It leaves starting and reverse it by leaving in between only

13. How do you extract every 3rd from a string starting from index

```
s = 'a1b2c3d4e5'
```

```
print(s[1::2])
```

output:

1 2 3 4 5

14. What is the output of the following, and explain how the negative step works negative step slices?

Explain start and stop behaviour when step is negative?

Ans: given:

```
s = 'python'
```

```
print(s[5:0:-1])
```

nohty

```
print(s[5::-1])
```

nohtyp

```
print(s[:0:-1])
```

nohty.

15. How do slice handle start and stop values that exceed string boundaries with negative step?

16. How do you extract the middle portion of a string dynamically, regardless of its length.

given:

```
s = 'abcdefghij'
```

```
print(s[2:9])
```

output

cd efghi

17. How do you use slicing to check if a string is a palindrome in a single line?

given:

```
s = 'racecar'
```

palindrome = If we reverse the order of the string or integer or character, the pronunciation and meaning will be same.

given: s = 'racecar'

```
print(s[::-1])
```

output:

racecar

18. What is the output of the following and why does it not raise an error?

19. How do you replace a portion of a string using slicing

Ans

```
s = 'Hello World'
```

```
print (s[:6] + 'python')
```

Output

Hello python

20. How does python's slice() object work, and how can you use it as a reusable slice?

Ans

```
s = 'a b c d e f g h i j'
```

```
my_slice = slice(1, 8, 2) # here (1, 8, 2) act as  
print (s[my_slice])
```

Output

b d f h

So, that it can
has to print the index
value from 1 to 8
by leaving 1 intermediate
value